

Basics of NLP: TF-IDF

TF-IDF: Measuring Word Importance

TF-IDF stands for Term Frequency-Inverse Document Frequency. It's a numerical statistic used to evaluate how important a word is to a specific document within a collection of documents (a corpus).

- Term Frequency (TF): Measures how often a word appears in a single document. A word that appears many times has a higher TF.
- Inverse Document Frequency (IDF): Measures how common or rare a word is across all documents. Common words (like "data") will have a low IDF score, while rare, specialized words will have a high IDF score.

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- TF-IDF Score: The final score is calculated by multiplying the TF and IDF.

$$TFIDF(t, d) = TF(t, d) \times \log \left(\frac{N}{DF(t)} \right)$$

Where:

t = term, d = document

$TF(t, d)$ = Measures how often term t appears in a document d .

$DF(t)$ = number of documents containing term t

N = total number of documents

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Step-by-Step TF-IDF Example

Sample Corpus

Let's say we have the following two documents:

Document 1 (D1): "Data science is fun"

Document 2 (D2): "Data science is powerful"

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Step-by-Step TF-IDF Example

Step 1: Preprocess the Text

Lowercase all words

Remove punctuation (if any)

Tokenize (split into words)

After preprocessing:

D1: ["data", "science", "is", "fun"]

D2: ["data", "science", "is", "powerful"]

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Step 2: Term Frequency (TF)

Calculate how often each term appears in a document.

D1: ["data", "science", "is", "fun"]

D2: ["data", "science", "is", "powerful"]

Term	TF in D1	TF in D2
data	1/4	1/4
science	1/4	1/4
is	1/4	1/4
fun	1/4	0
powerful	0	1/4

In this case, each word appears only once in each document, so raw term frequency = 1 (or 0 if absent).

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Step 3: Document Frequency (DF)

Number of documents in which each term appears.

D1: ["data", "science", "is", "fun"]

D2: ["data", "science", "is", "powerful"]

Term	TF in D1	TF in D2
data	1/4	1/4
science	1/4	1/4
is	1/4	1/4
fun	1/4	0
powerful	0	1/4

Term	DF
data	2
science	2
is	2
fun	1
powerful	1

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- **Step-by-Step TF-IDF Example**

Step 4: Inverse Document Frequency (IDF)

$$IDF(t) = \log_{10} \left(\frac{N}{DF(t)} \right)$$

Where N = 2 (total number of documents)

Term	TF in D1	TF in D2
data	1/4	1/4
science	1/4	1/4
is	1/4	1/4
fun	1/4	0
powerful	0	1/4

Term	DF
data	2
science	2
is	2
fun	1
powerful	1

Term	IDF	Value
data	$\log_{10}(2/2) = \log_{10}(1)$	0.00
science	$\log_{10}(2/2) = \log_{10}(1)$	0.00
is	$\log_{10}(2/2) = \log_{10}(1)$	0.00
fun	$\log_{10}(2/1) = \log_{10}(2)$	≈ 0.3010
powerful	$\log_{10}(2/1) = \log_{10}(2)$	≈ 0.3010

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Step-by-Step TF-IDF Example

Step 5: $TF \times IDF$

Now multiply TF with IDF for each term in each document:

Term	TF-IDF in D1	TF-IDF in D2
data	$1/4 \times 0 = 0$	$1/4 \times 0 = 0$
science	$1/4 \times 0 = 0$	$1/4 \times 0 = 0$
is	$1/4 \times 0 = 0$	$1/4 \times 0 = 0$
fun	$1/4 \times 0.301 = 0.075$	$0 \times 0.301 = 0$
powerful	$0 \times 0.301 = 0$	$1/4 \times 0.301 = 0.075$

Term	TF in D1	TF in D2
data	1/4	1/4
science	1/4	1/4
is	1/4	1/4
fun	1/4	0
powerful	0	1/4



Term	IDF Value
data	0.00
science	0.00
is	0.00
fun	≈ 0.3010
powerful	≈ 0.3010

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- Common words like "data", "science", and "is" have $\text{TF-IDF} = 0$ because they appear in all documents (not informative).
- Unique words like "fun" and "powerful" have higher TF-IDF values, making them more important for distinguishing between the documents.

Term	TF-IDF in D1	TF-IDF in D2
data	0	0
science	0	0
is	0	0
fun	0.075	0
powerful	0	0.075

Term	TF in D1	TF in D2
data	1/4	1/4
science	1/4	1/4
is	1/4	1/4
fun	1/4	0
powerful	0	1/4

Term	Value
data	0.00
science	0.00
is	0.00
fun	≈ 0.3010
powerful	≈ 0.3010